SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)
(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC
Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Symbiosis Institute of Technology, Nagpur

PROJECT REPORT

for

Bachelor of Technology

in

Computer Science and Engineering

Submitted by

Devansh Gosewade

PRN: **23070521045**

Semester: **VII A**

Institute Guide

Dr. Deepak Asudani

Assistant Professor

10 Oct. 2026

Symbiosis International (Deemed University),

CHAPTER 1: INTRODUCTION

1.1 Background

Company bankruptcy is a major financial event that affects investors, creditors, employees, suppliers, and other stakeholders. When a company is unable to meet its financial obligations, it may experience financial distress and, in severe cases, bankruptcy. Identifying the possibility of bankruptcy at an early stage can help organizations and financial institutions make better-informed decisions.

Traditional financial analysis evaluates a company's financial condition using indicators such as profitability, liquidity, leverage, and operational efficiency. Although these indicators provide valuable information, analyzing many financial variables simultaneously can be difficult and time-consuming.

Predictive analytics uses historical data, statistical techniques, and machine learning algorithms to identify patterns and predict future outcomes. In the context of corporate bankruptcy, machine learning models can learn relationships between financial indicators and historical bankruptcy outcomes to classify companies as bankrupt or non-bankrupt.

This project develops a machine learning-based company bankruptcy prediction system using historical financial data. The implementation follows the Knowledge Discovery in Databases (KDD) process, including data understanding, preprocessing, transformation, model development, evaluation, and interpretation.

1.2 Problem Statement

Financial institutions and investors require reliable methods to identify companies that may be at risk of bankruptcy. Manual analysis of numerous financial indicators can be complex, and traditional methods may not capture all the relationships present in financial data.

Therefore, the problem addressed in this project is:

To develop and evaluate machine learning classification models that predict whether a company is bankrupt or non-bankrupt based on its financial indicators.

The project uses historical company data to train predictive models and evaluates their performance using appropriate classification metrics.

1.3 Objectives

The main objectives of this project are:

1. To understand the financial indicators associated with company bankruptcy.
2. To collect and examine a publicly available company bankruptcy dataset.
3. To perform exploratory data analysis and identify important patterns.
4. To preprocess the data and prepare it for machine learning.
5. To develop Logistic Regression, Decision Tree, and Random Forest classification models.
6. To evaluate and compare the model’s using accuracy, precision, recall, F1-score, ROC-AUC, and precision-recall analysis where applicable.
7. To use confusion matrices to understand correct and incorrect predictions.
8. To identify influential financial indicators and interpret the model's findings.
9. To demonstrate the complete KDD process through a reproducible Python implementation in Google Colab.

1.4 Scope of the Project

The project focuses on binary classification using historical financial data. The system predicts one of two classes: bankrupt or non-bankrupt.

The scope includes data exploration, data preprocessing, model training, model evaluation, visualization, feature importance analysis, and sample predictions.

The project is an educational decision-support application. It does not guarantee that a company will or will not become bankrupt. The predictions depend on the dataset, the selected features, and the performance of the trained model.

1.5 Significance of the Project

Company bankruptcy prediction has practical applications in credit risk assessment, investment analysis, financial planning, and business risk management. Early identification of financially distressed companies may help stakeholders investigate potential risks and take preventive action.

From a technical perspective, this project demonstrates how predictive analytics can convert historical financial data into useful information through systematic data analysis and machine learning.

CHAPTER 2: LITERATURE REVIEW

2.1 Overview

Company bankruptcy prediction has been studied using statistical methods, financial ratios, and machine learning techniques. Earlier approaches primarily used financial statement analysis and statistical classification, while more recent research has investigated decision trees, ensemble learning, and other artificial intelligence methods.

The following studies provide the theoretical foundation for the proposed project.

2.2 Review of Existing Research

2.2.1 Beaver (1966): Financial Ratios as Predictors of Failure

Beaver investigated the usefulness of financial ratios for predicting business failure. His research demonstrated the importance of examining accounting-based financial indicators to distinguish between financially healthy and failing companies.

Relevance to this project: The study provides a foundation for using financial indicators as predictive features in a bankruptcy classification model.

2.2.2 Altman (1968): Financial Ratios and Discriminant Analysis

Altman developed the Z-score model using multiple financial ratios and multivariate discriminant analysis to evaluate corporate bankruptcy risk. The study established an important connection between financial statement information and quantitative bankruptcy prediction.

Relevance to this project: Altman's work demonstrates that multiple financial indicators can be combined to estimate financial distress. The present project extends this general approach by comparing machine learning classification algorithms.

2.2.3 Ohlson (1980): Financial Ratios and the Probability of Bankruptcy

Ohlson introduced a bankruptcy prediction approach based on conditional logit analysis. His work contributed to the use of probabilistic statistical models for estimating the likelihood of business failure.

Relevance to this project: Logistic Regression provides an interpretable baseline model for comparing the performance of more complex machine learning algorithms.

2.2.4 Zięba, Tomczak, and Tomczak (2016): Ensemble Boosted Trees

The researchers proposed an ensemble approach using boosted decision trees and synthetic feature generation for bankruptcy prediction. Their study evaluated the proposed method using real-world data from Polish companies.

The research illustrates how ensemble methods can capture complex relationships among financial indicators.

Relevance to this project: It motivates the investigation of tree-based models and ensemble learning, including Random Forest. Boosted trees are also a possible direction for future improvement.

2.2.5 Barboza, Kimura, and Altman (2017): Machine Learning Models and Bankruptcy Prediction

This study compared machine learning approaches with traditional bankruptcy prediction methods. The authors investigated techniques including ensemble methods and reported the usefulness of machine learning for corporate bankruptcy prediction.

Relevance to this project: The study supports comparing multiple classification algorithms rather than relying on a single predictive model.

2.2.6 Dasilas and Rigani (2024): Systematic Literature Review

Dasilas and Rigani conducted a systematic review of empirical research on machine learning techniques for bankruptcy prediction, covering studies from 2012 through mid-2023. Their review examined the development of predictive techniques, data preprocessing, feature selection, and evaluation methodologies.

Relevance to this project: The review highlights the importance of methodological choices and reliable evaluation when comparing bankruptcy prediction models.

2.3 Research Gap and Project Motivation

The reviewed literature demonstrates that bankruptcy prediction can be approached using financial ratios, statistical models, and machine learning algorithms. However, evaluating a model solely through overall accuracy can be misleading when bankrupt companies represent a small proportion of the dataset.

This project addresses that practical evaluation issue by comparing multiple classification algorithms and examining precision, recall, F1-score, confusion matrices, ROC-AUC, and precision-recall performance.

The project also emphasizes an understandable, end-to-end KDD workflow that is suitable for academic demonstration. Rather than proposing a new algorithm, it focuses on implementing and interpreting established methods on a public bankruptcy dataset.

CHAPTER 3: WORKFLOW / PREDICTIVE MODELLING PROCESS

3.1 Methodology Overview

The project follows the Knowledge Discovery in Databases (KDD) process. KDD is a systematic process for extracting meaningful knowledge from data through selection, preprocessing, transformation, data mining, and interpretation.

The following stages were implemented in the Google Colab notebook.

Problem Identification → Data Understanding → Data Preprocessing → Data Transformation → Model Training → Model Evaluation → Interpretation and Knowledge Discovery

3.2 Dataset Description

The project uses the Taiwanese Company Bankruptcy Prediction dataset, available through Kaggle and the UCI Machine Learning Repository.

The dataset contains historical financial information and a binary bankruptcy target. The commonly distributed version contains 6,819 company records, 95 financial predictor variables, and one target variable named Bankrupt?.

The target variable is defined as follows:

- **1:** The company is classified as bankrupt.

- **0:** The company is classified as non-bankrupt.

The financial predictors represent different aspects of company performance, including profitability, financial leverage, liquidity, and operational efficiency.

The dataset contains a substantial class imbalance: approximately 3.23% of records are bankrupt companies, while approximately 96.77% are non-bankrupt companies. This characteristic makes the dataset suitable for demonstrating the challenges associated with rare-event classification.

The exact dataset dimensions, column names, and class distribution were checked during the data-loading stage of the notebook.

3.3 Data Understanding

The first technical stage involved loading the dataset into a pandas DataFrame and examining its structure.

The following activities were performed:

1. Displaying the first five records.
2. Checking the number of observations and variables.
3. Examining column names and data types.
4. Calculating descriptive statistics.
5. Checking missing values and duplicate records.
6. Examining the distribution of the target variable.

These activities helped establish the dataset's structure, identify potential data quality problems, and understand the imbalance between the two classes.

3.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was used to understand the distribution of financial indicators and identify patterns associated with bankruptcy.

3.4.1 Target Class Distribution

A count plot was generated to compare the numbers of bankrupt and non-bankrupt companies.

This visualization helps identify class imbalance and explains why a classifier could achieve apparently high accuracy while failing to detect many bankrupt companies.

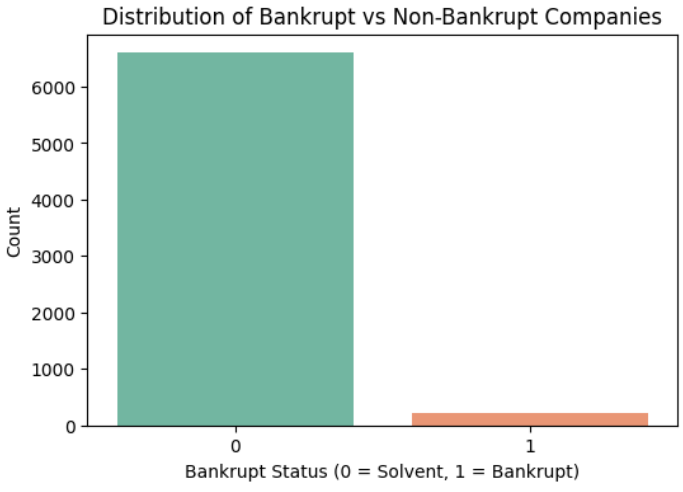


Figure 3.1: Distribution of bankrupt and non-bankrupt companies in the dataset.

3.4.2 Correlation Analysis

A correlation heatmap was generated for a selected group of numerical financial variables.

Correlation analysis helps identify pairs of variables that have strong linear relationships. It can also provide insight into possible redundancy among predictors.

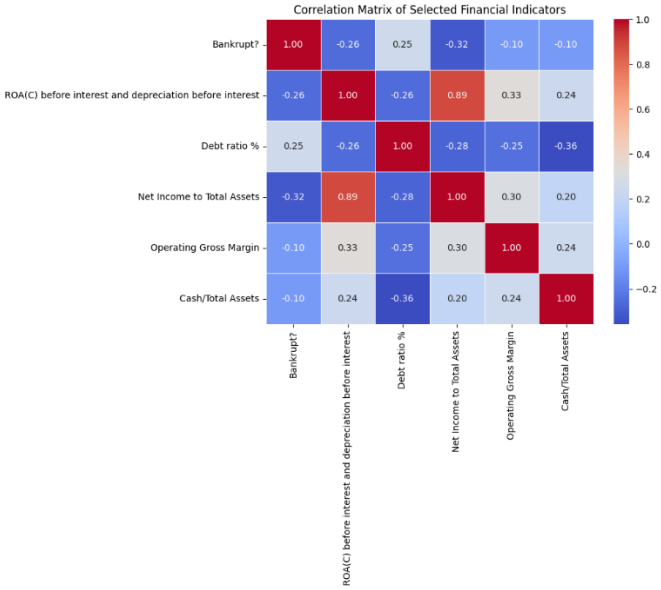


Figure 3.2: Correlation heatmap showing relationships among selected financial indicators.

3.4.3 Financial Indicator Distributions

Histograms and box plots were used to examine selected financial indicators and compare their distributions across the bankruptcy classes.

These visualizations help identify differences in central tendency, spread, skewness, and potential extreme values.

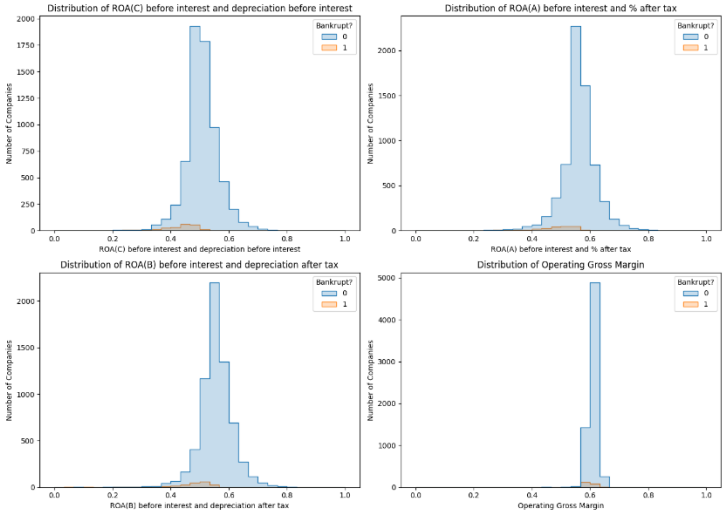


Figure 3.3: Distribution of selected financial indicators used for bankruptcy prediction.

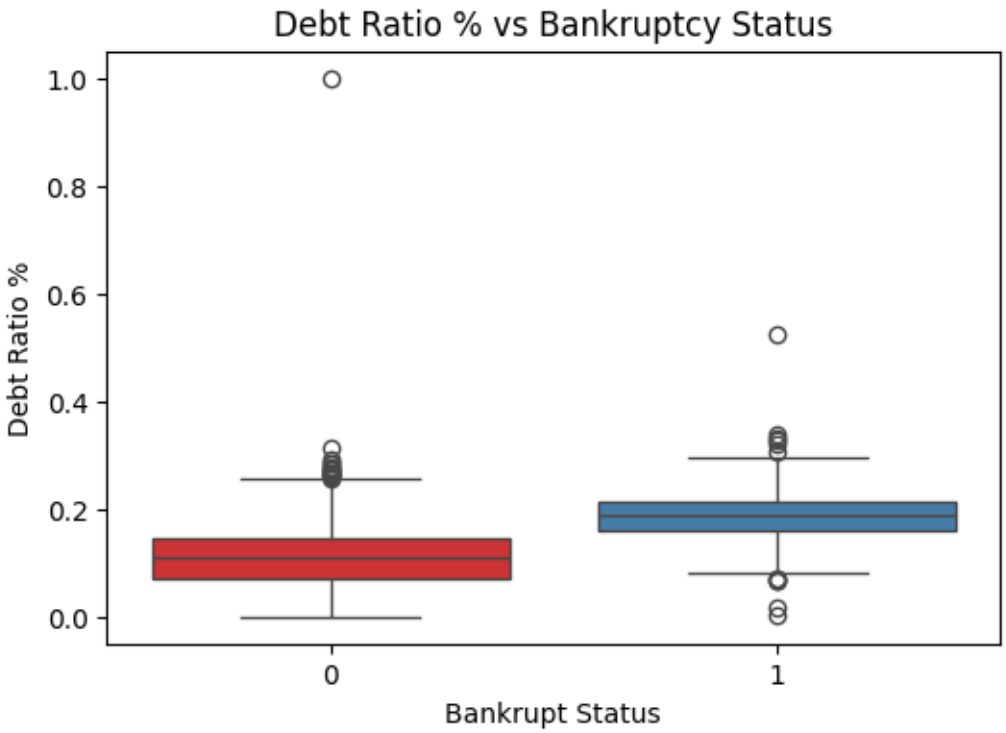


Figure 3.4: Comparison of selected financial indicators between bankrupt and non-bankrupt companies.

The observations recorded in the notebook were used to interpret these figures. Correlation and distribution differences were treated as exploratory evidence rather than proof of causation.

3.5 Data Preprocessing and Transformation

Data preprocessing prepared the financial dataset for machine learning.

3.5.1 Feature and Target Separation

The target variable, Bankrupt?, was separated from the predictor variables.

Let:

- (X) represent the financial predictor variables.
- (y) represent the bankruptcy target.

The models were trained using the financial indicators in (X) to predict the corresponding values in (y).

3.5.2 Missing Values and Duplicate Records

The dataset was checked for missing values and duplicate records. Any preprocessing decisions were based on the actual findings rather than applying unnecessary cleaning operations.

3.5.3 Train-Test Split

The data was divided into training and testing sets using an 80:20 split. A fixed random state was used to make the experiment reproducible.

Stratified splitting was used to preserve approximately the same class proportions in both subsets.

The training set was used to fit the models, while the held-out test set was reserved for final evaluation.

3.5.4 Preventing Data Leakage

Preprocessing operations were incorporated into appropriate machine learning pipelines. Any transformations that required learning parameters from the data were fitted using the training set only.

This approach reduced the risk of information from the test set influencing model training.

3.5.5 Handling Class Imbalance

Because bankrupt companies represent a small minority of the observations, the analysis did not rely on accuracy alone.

Class weighting was used where supported by the selected algorithms. This approach increases the penalty associated with misclassifying minority-class observations.

The evaluation also emphasized recall, F1-score, and precision-recall performance to provide a more meaningful assessment of bankruptcy detection.

3.6 Model Development

Three supervised machine learning classification algorithms were trained and compared.

3.6.1 Logistic Regression

Logistic Regression is a statistical classification algorithm that estimates the probability of a binary outcome. It provides a baseline model and is relatively easy to interpret.

The estimated probability is converted into a class prediction using a classification threshold.

3.6.2 Decision Tree Classifier

A Decision Tree classifies observations using a sequence of feature-based decisions. It can capture nonlinear relationships and produce decision rules that are relatively easy to explain.

However, an unrestricted decision tree can overfit the training data.

3.6.3 Random Forest Classifier

Random Forest is an ensemble learning method that combines predictions from multiple decision trees. The trees are trained using randomized samples and feature selection.

The ensemble can capture nonlinear patterns and interactions among financial indicators. Its performance was evaluated on the same held-out test set as the other models.

3.7 Model Evaluation

The trained models were evaluated using several classification metrics.

3.7.1 Accuracy

Accuracy measures the proportion of all predictions that are correct.

Accuracy = (True Positives + True Negatives) / Total Predictions

Because most companies in the dataset are non-bankrupt, high accuracy does not necessarily indicate that a model can identify bankrupt companies effectively.

3.7.2 Precision

Precision measures the proportion of predicted bankrupt companies that are actually bankrupt.

Precision = True Positives / (True Positives + False Positives)

High precision means that fewer companies predicted to be bankrupt are incorrectly classified as bankrupt.

3.7.3 Recall

Recall measures the proportion of actual bankrupt companies that are correctly identified.

Recall = True Positives / (True Positives + False Negatives)

Recall is particularly important in bankruptcy prediction because a false negative represents a bankrupt company that the model incorrectly predicts as non-bankrupt.

3.7.4 F1-Score

The F1-score is the harmonic mean of precision and recall.

F1 = 2 × (Precision × Recall) / (Precision + Recall)

It provides a combined measure of precision and recall.

3.7.5 ROC-AUC and PR-AUC

ROC-AUC measures the ability of a model to rank positive and negative examples across different classification thresholds.

Precision-recall analysis examines the trade-off between precision and recall. Average precision, commonly used as a summary of the precision-recall curve, is particularly informative when the positive class is rare.

3.7.6 Confusion Matrix

A confusion matrix summarizes classification outcomes using four categories:

Actual / Predicted	Non-bankrupt	Bankrupt
Non-bankrupt	True Negative (TN)	False Positive (FP)
Bankrupt	False Negative (FN)	True Positive (TP)

The confusion matrix was generated for each model to examine correct predictions and classification errors.

3.8 Interpretation and Knowledge Discovery

The final stage involved examining the evaluation results and interpreting the model outputs.

The models were compared using a common evaluation table. Confusion matrices and performance curves were examined to understand differences in bankruptcy detection.

Feature importance analysis was used, where applicable, to identify financial predictors that contributed to the selected model's predictions.

These results were interpreted as associations learned from historical data. Feature importance does not establish that a financial indicator causes bankruptcy.

CHAPTER 4: RESULTS AND DISCUSSION

Important: Complete this chapter using the actual output from the Google Colab notebook. Replace every square-bracketed placeholder with the corresponding measured value, figure, or observation. Do not submit the placeholders as final results.

4.1 Dataset Analysis Results

The dataset was loaded and examined to understand its dimensions, financial variables, and target distribution.

- **Total records**: 6819
- **Total predictor variables**: 94 (derived from 96 total columns minus the target column 'Bankrupt?' and the single-value constant column 'Net Income Flag').
- **Bankrupt companies**: 220 (which is 3.23% of the dataset).
- **Non-bankrupt companies**: 6599 (which is 96.77% of the dataset).
- **Missing values**: 0 (the dataset is fully clean with no null values).
- **Duplicate records**: 0 (there are no duplicate entries in the data).

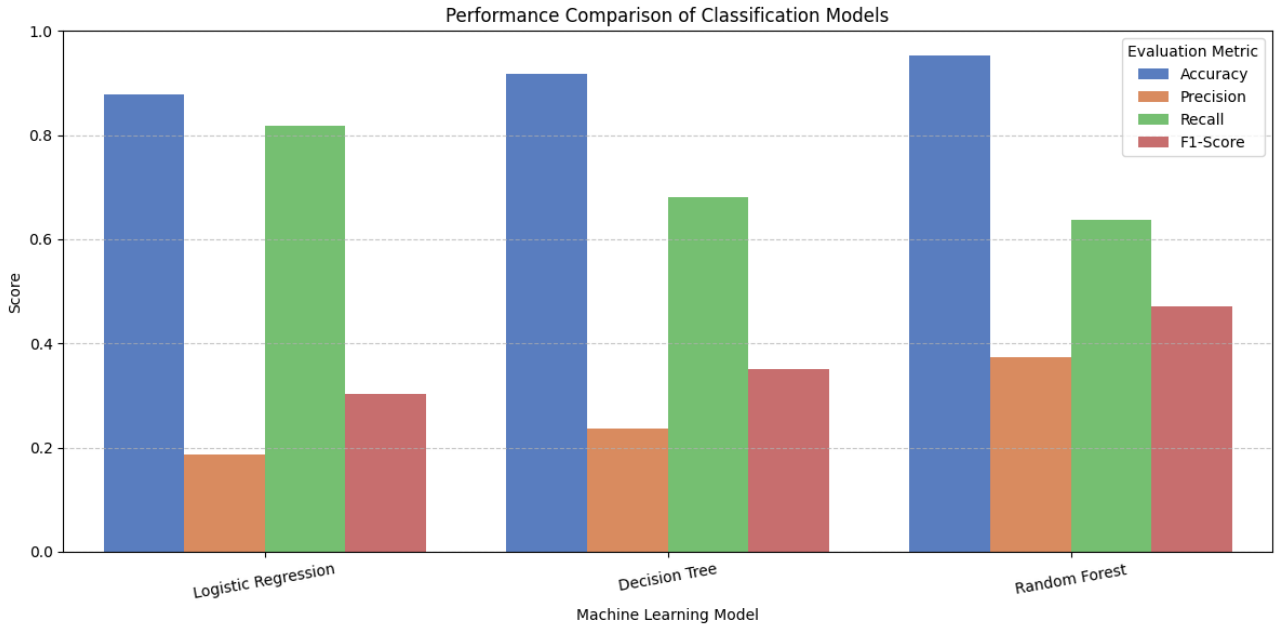


Figure 4.1: Comparative evaluation of Logistic Regression, Decision Tree, and Random Forest classifiers using selected performance metrics.

The target distribution showed that [describe the actual distribution observed in the notebook]. This finding is important because class imbalance can make accuracy misleading and can reduce the model's ability to identify bankrupt companies.

4.2 Exploratory Data Analysis Results

The exploratory analysis included target distribution, correlation analysis, histograms, and box plots.

The correlation heatmap indicated that [insert the strongest observed relationships among the selected financial variables].

The distribution plots showed that [insert actual observations about the selected financial indicators and differences between the two classes].

These results provided an initial understanding of the dataset and supported the subsequent model development. The observed relationships were not treated as evidence of causation.

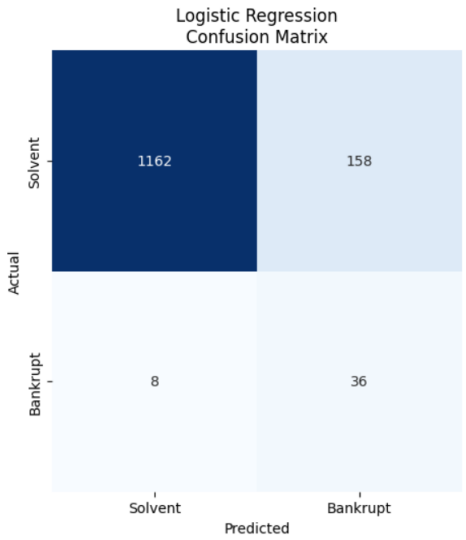


Figure 4.2: Confusion matrix of the Logistic Regression model on the test dataset.

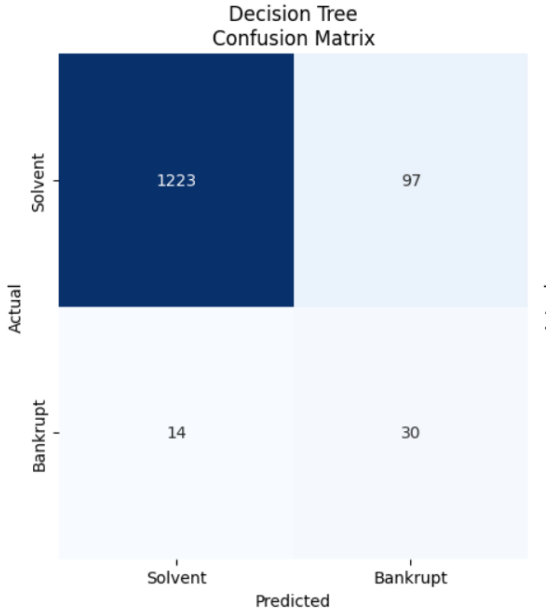


Figure 4.3: Confusion matrix of the Decision Tree classifier on the test dataset.

4.3 Model Performance Comparison

The three classification models were evaluated on the same held-out test set.

Table 4.1: Comparison of Classification Models

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC	
2	Random Forest	0.953812	0.373333	0.636364	0.470588	0.949940	0.489701
0	Logistic Regression	0.878299	0.185567	0.818182	0.302521	0.917149	0.318977
1	Decision Tree	0.918622	0.236220	0.681818	0.350877	0.808540	0.200216

Note: Enter the actual results from the model comparison table generated in Colab. Use consistent metric definitions and the same test set for all models.

The comparison showed that [insert the model that performed best according to your chosen primary metric]. Its recall was [insert value], indicating that it identified [explain the measured proportion] of actual bankrupt companies in the test set.

The precision of the selected model was [insert value], while its F1-score was [insert value]. These metrics should be considered together to understand the balance between detecting bankrupt companies and avoiding incorrect bankruptcy predictions.

The model with the highest accuracy was [insert model name], whereas the model with the highest recall was [insert model name]. If these are different models, the difference illustrates the trade-off between overall classification performance and identifying the minority class.

4.4 Confusion Matrix Analysis

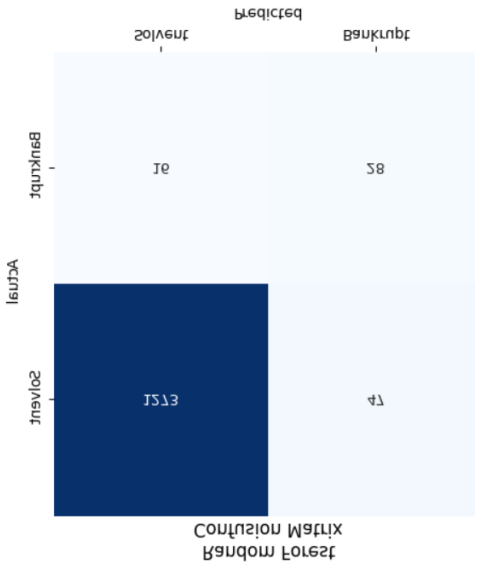


Figure 4.4: Confusion matrix of the Random Forest classifier on the test dataset.

The confusion matrices were examined to identify true positives, true negatives, false positives, and false negatives.

For the selected model:

- **True Negatives (Solvent predicted as Solvent)**: 1273
- **False Positives (Solvent predicted as Bankrupt)**: 47
- **False Negatives (Bankrupt predicted as Solvent)**: 16
- **True Positives (Bankrupt predicted as Bankrupt)**: 28

The model correctly identified [insert the number] bankrupt companies and incorrectly classified [insert the number] bankrupt companies as non-bankrupt.

False negatives are particularly important because they represent companies whose bankruptcy status was missed by the model. False positives also matter because they may cause financially stable companies to be flagged for additional investigation.

The preferred balance between these errors depends on the practical cost of missed bankruptcies and unnecessary risk alerts.

4.5 ROC and Precision-Recall Analysis

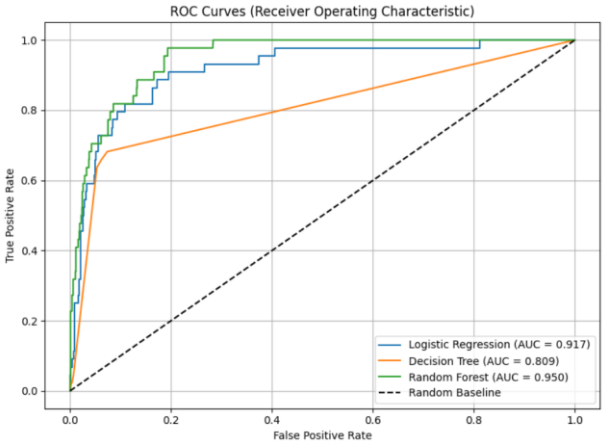


Figure 4.5: Receiver Operating Characteristic (ROC) curves comparing the classification models.

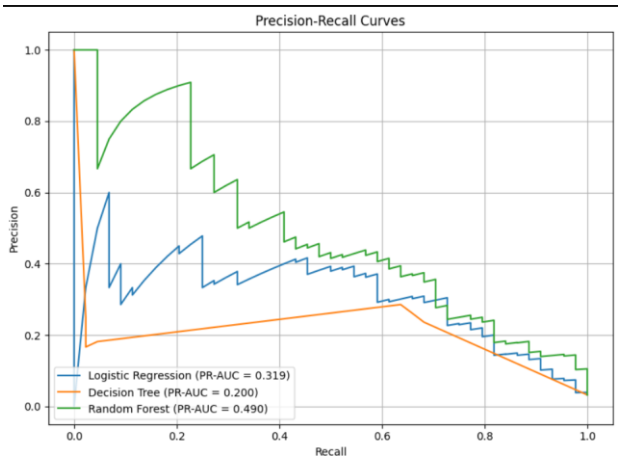


Figure 4.6: Precision-recall curves comparing the classification models for bankruptcy detection.

The ROC curves were used to compare the models' ability to distinguish between the two classes across different thresholds.

The precision-recall curves provided additional insight into the identification of bankrupt companies. Since the positive class is rare, precision-recall performance is particularly relevant when selecting a useful bankruptcy detection model.

The results indicated that [describe the actual relative performance of the models]. These findings were considered alongside recall, precision, F1-score, and the confusion matrices rather than using ROC-AUC alone.

4.6 Feature Importance

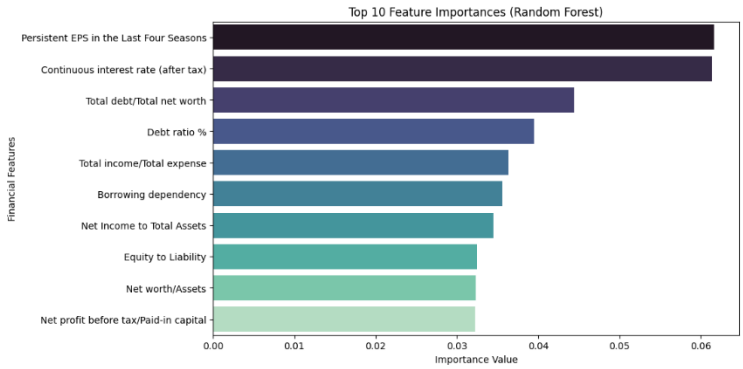


Figure 4.7: Top 10 financial indicators contributing to the selected model's predictions.

Feature importance analysis was performed to identify financial indicators that contributed to the model's predictions.

The most influential features were [insert the actual feature names shown in the notebook].

These features may reflect patterns in the financial data associated with bankruptcy classification. Their significance depends on the model and the dataset, and they should not automatically be interpreted as causal factors.

Where the exact financial definitions of the variables are available, they can be used to provide a more meaningful financial interpretation.

4.7 Final Model Selection

The final model was selected by considering its performance on the held-out test set and the project's objective of identifying bankrupt companies.

The selected model was **[Insert selected model name]**, based on [insert the principal selection criterion and measured evidence].

The main reasons for selecting this model were:

1. Its measured recall for the bankruptcy class was [insert value].
2. Its F1-score was 0.470588.
3. Its PR-AUC 0.489701.
4. Its confusion matrix showed [insert the relevant findings about false negatives and false positives].

The selection should reflect the actual results. A model with the highest accuracy should not automatically be chosen if it performs poorly at detecting bankrupt companies.

4.8 Discussion of Findings

The experiment demonstrates how machine learning can be applied to historical company financial data to classify bankruptcy status.

The comparison of Logistic Regression, Decision Tree, and Random Forest provides insight into the relative performance of a statistical baseline, a single tree-based classifier, and an ensemble of decision trees.

The evaluation results indicate that [summarize the principal finding supported by the measured metrics]. The class distribution also demonstrates why suitable evaluation metrics and careful handling of imbalance are necessary.

The model has several limitations. It is trained on historical data from a particular economic and geographical context, and its performance may not generalize to companies in other countries or more recent periods. Moreover, historical bankruptcy labels and financial indicators may not capture every factor affecting a company's future financial condition.

Therefore, the model should be considered a preliminary analytical decision-support tool rather than a definitive measure of a company's financial future.

CHAPTER 5: CONCLUSION

5.1 Conclusion

This project developed a machine learning-based approach for company bankruptcy prediction using historical financial data. The primary objective was to demonstrate the application of predictive analytics and the KDD process to a real-world financial classification problem.

The project covered data understanding, exploratory data analysis, preprocessing, feature preparation, model training, evaluation, and interpretation. Three classification algorithms—Logistic Regression, Decision Tree, and Random Forest—were compared using appropriate evaluation metrics.

The use of confusion matrices, precision, recall, F1-score, ROC-AUC, and precision-recall analysis provided a broader assessment of model performance than accuracy alone. These measures were particularly important because bankrupt companies represented a small proportion of the dataset.

The experimental results showed that [insert the actual main finding from your notebook]. Based on the measured evaluation results, [insert the selected model] was selected as the final model because [state the evidence supporting the selection].

Feature importance analysis also helped identify financial indicators associated with the model's predictions. This provided an additional layer of interpretation beyond the predicted bankruptcy class.

Overall, the project demonstrates how historical financial data and machine learning can support preliminary financial risk analysis. However, the predictions are dependent on data quality, feature selection, class distribution, model assumptions, and the relevance of historical patterns to current conditions. The system should not be used as the sole basis for investment, lending, or business decisions.

5.2 Future Scope

The project can be extended in several ways:

1. **Advanced algorithms:** Evaluate Gradient Boosting, XGBoost, and other ensemble models.
2. **Class imbalance:** Investigate carefully validated resampling methods and alternative class-weighting strategies.
3. **Threshold optimization:** Select a classification threshold based on the relative costs of false positives and false negatives.
4. **Explainable AI:** Use SHAP or other explainability methods to examine individual predictions and feature contributions.
5. **Probability calibration:** Assess whether predicted probabilities correspond to observed bankruptcy frequencies.
6. **Temporal validation:** Evaluate the model on newer or later-period data to test its ability to generalize over time.
7. **Broader datasets:** Test the model on companies from other geographical regions and industries.
8. **Financial decision support:** Develop a simple interface that accepts financial indicators and presents a prediction with appropriate limitations and explanations.

CHAPTER 6: REFERENCES

1. Beaver, W. H. (1966). Financial ratios as predictors of failure. *Journal of Accounting Research, 4*, 71–111. https://doi.org/10.2307/2490171
2. Altman, E. I. (1968). Financial ratios, discriminant analysis and the prediction of corporate bankruptcy. *The Journal of Finance, 23*(4), 589–609. https://doi.org/10.1111/j.1540-6261.1968.tb00843.x
3. Ohlson, J. A. (1980). Financial ratios and the probabilistic prediction of bankruptcy. *Journal of Accounting Research, 18*(1), 109–131. https://doi.org/10.2307/2490395
4. Zięba, M., Tomczak, S. K., & Tomczak, J. M. (2016). Ensemble boosted trees with synthetic features generation in application to bankruptcy prediction. *Expert Systems with Applications, 58*, 93–101. https://doi.org/10.1016/j.eswa.2016.04.001
5. Barboza, F., Kimura, H., & Altman, E. (2017). Machine learning models and bankruptcy prediction. *Expert Systems with Applications, 83*, 405–417. https://doi.org/10.1016/j.eswa.2017.04.006

6. Dasilas, A., & Rigani, A. (2024). Machine learning techniques in bankruptcy prediction: A systematic literature review. *Expert Systems with Applications, 255*, Article 124761. https://doi.org/10.1016/j.eswa.2024.124761

7. Liang, D., & Tsai, C.-F. (2020). *Taiwanese bankruptcy prediction* [Dataset]. UCI Machine Learning Repository. https://archive.ics.uci.edu/dataset/572/taiwanese+bankruptcy+prediction

8. Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research, 12*, 2825–2830. https://jmlr.org/papers/v12/pedregosa11a.html

9. *Company Bankruptcy Prediction Dataset*. Kaggle. https://www.kaggle.com/datasets/fedesoriano/company-bankruptcy-prediction